Interpretation:-

1. D: Assign the Domain.
2. F: Assign the Constants from domain
3. P: Assign a specific relation from the domain.

Step-1 Take D: { Set of all real number }

Relation
Step 2:-
- q (x, y): x is greater than y
- K (x): x is even number.
- r (x, y, z): Greatest number of x, y, z.

→ Remove ∀, ∃. { Skolemization function }

Step-3

↓ Sentences. $\{(F_1 \cap F_2) \to G\}$

F:- → CNF $(C_1 \cap C_2 \cap \dots C_n)$.

Step-4:- G, $\neg G$.

Step-5: Add $\neg G$, $(C_1 \cap C_2 \cap C_3 \cap \dots C_n \cap \neg G)$

↓ Simplify & Understand

A formula is said to valid if it is true for all interpretations.

Propotional logic:-

Ashe is elected as VP $\rightarrow a$
Rejat is chosen as Grace $\rightarrow b$
Bharti is chosen as Treasurer $\rightarrow c$

$\neg b \rightarrow$ Given

$\neg a \rightarrow C_1$

$(C_1 \cap C_2 \cap C_3) \rightarrow C_1$ $(\neg a)$

$(C_1 \cap C_2 \cap C_3 \cap a)$ Given $\rightarrow$ False

$p \rightarrow q$

$\neg p \vee q$

$$(a \rightarrow (b \cap c)) = (\neg a \vee (b \cap c))$$

$$= (\neg a \vee b) \cap \neg a \vee c) = (\neg a \vee b) \cap (\neg a \vee c)$$

$$= \neg b$$

$(\neg a \vee \neg b)$ – C_1

$(\neg a \vee \neg c)$ – C_2

$(\neg b)$ – C_3

$\boxed{a}$ – C_4

$\boxed{\neg a}$

$(C_1 \cap C_2 \cap C_3 \cap C_4 \cap \neg C_1) \longrightarrow$ False

Resolution Refutation for Propositional Logic

To prove validity of

F = ((F1 ∧ F2 ∧ … ∧ Fn) → G)

we shall attempt to prove that

~F = (F1 ∧ F2 ∧ … ∧ Fn ∧ ~G)

is unsatisfiable

Steps for Proof by Resolution Refutation:

1. **Convert of Clausal Form / Conjunctive Normal Form (CNF, Product of Sums).**
2. **Generate new clauses using the resolution rule.**
3. **At the end, either False will be derived if the formula ~F is unsatisfiable implying F is valid.**

If Asha is elected VP then Rajat is chosen as G-Sec and Bharati is chosen as Treasurer. Rajat is not chosen as G-Sec. Therefore Asha is not elected VP.

F1: (a → (b ∧ c)) = (~a V b) ∧ (~a V c)

F2: ~b, G: ~a, ~G: a

Clauses of Clause Form: ~F = (C1 ∧ C2 ∧ C3 ∧ C4)

where: C1: (~a V b)

C2: (~a V c)

C3: ~b

C4: a

To prove that ~F is False

Let C1 = a V b and C2 = ~a V c

then a new clause C3 = b V c can be derived.

(Proof by showing that ((C1 ∧ C2) → C3) is a valid formula).

To prove unsatisfiability use the Resolution Rule repeatedly to reach a situation where we have two contradictory clauses of the form C1 = a and C2 = ~a from which False can be derived.

If the propositional formula is satisfiable then we will not reach a contradiction and eventually no new clauses will be derivable.

For propositional logic the procedure terminates.

Resolution Rule is Sound and Complete

Applying Resolution Refutation

Let C1 = a V b and C2 = ~a V c

then a new clause C3 = b V c can be derived.

(Proof by showing that ((C1 ∧ C2) → C3) is a valid formula).

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F1: (a → (b ∧ c)) = (~a V b) ∧ (~a V c)

F2: ~b

G: ~a

~G: a

Clauses of Clause Form: ~F = (C1 ∧ C2 ∧ C3 ∧ C4)

where: C1: (~a V b)

C2: (~a V c)

C3: ~b

C4: a

To prove that ~F is False

New Clauses Derived

C5: ~a (Using C1 and C3)

C6: False (using C4 and C5)

Example

Let C1 = a V b and C2 = ~a V c then a new clause C3 = b V c can be derived.

*(Proof by showing that ((C1 ∧ C2) → **C3) is a valid formula).***

To prove unsatisfiability use the Resolution Rule repeatedly to reach a situation where we have two contradictory clauses of the form C1 = a and C2 = ~a from which False can be derived.

If the propositional formula is satisfiable then we will not reach a contradiction and eventually no new clauses will be derivable.

For propositional logic the procedure terminates.

Resolution Rule is Sound and Complete

Rajesh either took the bus or came by cycle to class. If he came by cycle or walked to class he arrived late. Rajesh did not arrive late. Therefore he took the bus to class.

$F_1 :- (bus \wedge \neg cycle) \vee (\neg bus \wedge cycle)$.

$F_2 :- (Cycle \vee Walk) \rightarrow Late$.

$F_3 :- \neg late$

$G :- bus$; $\sim G :- \neg bus$.

$F :- (F_1 \cap F_2 \cap F_3 \cap \dots \cap F_n \cap \sim G)$

$F_1 :- (bus \cap \neg cycle) \vee (\neg bus \cap cycle)$

$:- \underbrace{(bus \vee cycle)}_{C_1} \wedge \underbrace{(\neg cycle \vee \neg bus)}_{C_2}$.

$F_2 :- (Cycle \vee walk) \rightarrow late$ $\qquad P \rightarrow q = \neg p \vee q$.

$(\neg(Cycle \vee walk) \vee late)$

$(\neg Cycle \wedge \neg walk) \vee late)$. C_4

$F_2 :- (\neg Cycle \vee late) \wedge (\neg walk \vee late)$.

C_3

$C_5 :- \sim late$ $\quad C_6 :- \neg bus$ $\qquad C_6 \& C_4$

$C_7 :- Cycle$

$C_3 \& C_7 :- late$

False $\vdash \square$

Implicit Quantifier notation

Prefix universally quantified variables with a "?". Replace existentially quantified variables not in the scope of a universal quantified with a *Skolem constant* (named after the mathematician Thoralf Skolem)

Drop the quantifiers

Universal → ∀

Existential → ∃

Kumar is a student

All students are smart

Student(x), Smart(x).

G:- $\boxed{\forall x}$ (Student(x) → Smart(x)).

$(p(x) \to q(x))$.

F2:- Student(Kumar)

F1:- Student(?Kumar) → Smart(?Kumar).

$p(x)$.

Student(Kumar)

G:- Smart(Kumar)

① $p(?K) \to q(?K)$

② $p(?K)$.

$q(K)$

$\alpha \to \beta$

α

β

MPP: an example

Thus if

α = (Sport(tennis) $\wedge$ Likes(Alice, tennis))

$\beta \supset \delta$ = (Sport(?y) $\wedge$ Likes(?x, ?y)) $\supset$ Watches(?x, ?y)

then α unifies with β

with the substitution θ = {<?x, Alice>, <?y, tennis>}

and one can infer

$\delta\theta$ = Watches(?x, ?y)θ = Watches(Alice, tennis)

Resolution Refutation for Predicate Logic

Given a formula F which we wish to check for validity, we first check if there are any free variables. We then quantify all free variables universally.

Create F' = ~F and check for unsatisfiability of F'

STEPS:

Conversion to Clausal (CNF) Form:

- Handling of Variables and Quantifiers, Ground Instances

Applying the Resolution Rule:

- Concept of Unification
- Principle of Most General Unifier (mgu)
- Repeated application of Resolution Rule using mgu

CONVERSION TO CLAUSAL FORM IN PREDICATE LOGIC

1. Remove implications and other Boolean symbols converting to equivalent forms using ~, V, Λ
2. Move negates (~) inwards as close as possible
3. Standardize (Rename) variables to make them unambiguous
4. Remove Existential Quantifiers by an appropriate new function /constant symbol taking into account the variables dependent on the quantifier (Skolemization)
5. Drop Universal Quantifiers
6. Distribute V over Λ and convert to CNF

- Anyone who likes animal is liked by ~~everyone~~ Someone

Symbols:- Animal,

Predicate: likes (x, y) – x likes y.

Animal (x)

$$\forall x \{ (\forall y \; \text{Animal}(y) \rightarrow \text{likes}(x, y)) \} \Rightarrow$$

$$\rightarrow \exists y \; (\text{likes}(y, x)).$$

$$\forall x \{ \neg \forall y \; (\text{Animal}(y) \rightarrow \text{likes}(x, y)) \} \vee \exists y \; (\text{likes}(y, x)).$$

$$\forall x \{ \neg \forall y \; (\neg \text{Animal}(y) \vee \text{likes}(x, y)) \} \vee \exists y \; (\text{likes}(y, x)).$$

$$\forall x \; (\exists y \; (\text{Animal}(y) \wedge \neg \text{likes}(x, y))) \vee \exists y \; (\text{likes}(y, x)).$$

Standardize the variable : K'

$$\forall x\,(\exists y\,(\text{Animal}(y) \wedge \neg \text{likes}(x,y))) \vee \exists K\,(\text{likes}(K,x)).$$

Remove Existential Quantifiers

Skolemization Functions $\begin{cases} y - z(x). \\ K - w(x). \end{cases}$

$$\boxed{\forall x}\,(\text{Animal}(z(x)) \wedge \neg \text{likes}(x, z(x))) \vee \text{likes}(w(x), x)$$

Drop Universal Quantifiers:

$$(\text{Animal}((z(x)) \wedge \neg \text{likes}(x, z(x))) \vee \text{likes}(w(x), x).$$

Distribution fact here:

$$\underbrace{\{\text{Animal}(z(x) \vee \text{likes}(w(x), x)\}}_{C_1} \wedge \underbrace{\{\neg \text{likes}(x, z(x)) \vee \text{likes}(w(x), x)\}}_{C_2}.$$

CNF : $\neg$ Proposition

$$(C_1 \cap C_2 \cap \sim G) \rightarrow \text{False}.$$

Resolution Refutation for Predicate Logic

Given a formula F which we wish to check for validity, we first check if there are any free variables. We then quantify all free variables universally.

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STEPS:

Conversion to Clausal (CNF) Form:

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- Repeated application of Resolution Rule using mgu

F1: ∀x(goes(Mary, x) → goes(Lamb, x))

F2: goes(Mary, School)

G: goes(Lamb, School)

To prove: (F1 ∧ F2) → G) is valid

CONVERSION TO CLAUSAL FORM IN PREDICATE LOGIC

1. Remove implications and other Boolean symbols converting to equivalent forms using ~, V, ∧
2. Move negates (~) inwards as close as possible
3. Standardize (Rename) variables to make them unambiguous
4. Remove Existential Quantifiers by an appropriate new function /constant symbol taking into account the variables dependent on the quantifier (Skolemization)
5. Drop Universal Quantifiers
6. Distribute V over ∧ and convert to CNF

Examples

F1: ∀x(**contractor(x)** → **~dependable(x)**)

F2: ∃x(engineer(x) ∧ contractor(x))

G: ∃x(engineer(x) ∧ ~dependable(x))

F1: ∀x(dancer(x) → graceful (x))

F2: student(Ayesha), F3: dancer(Ayesha)

G: ∃x(student (x) ∧ graceful(x))

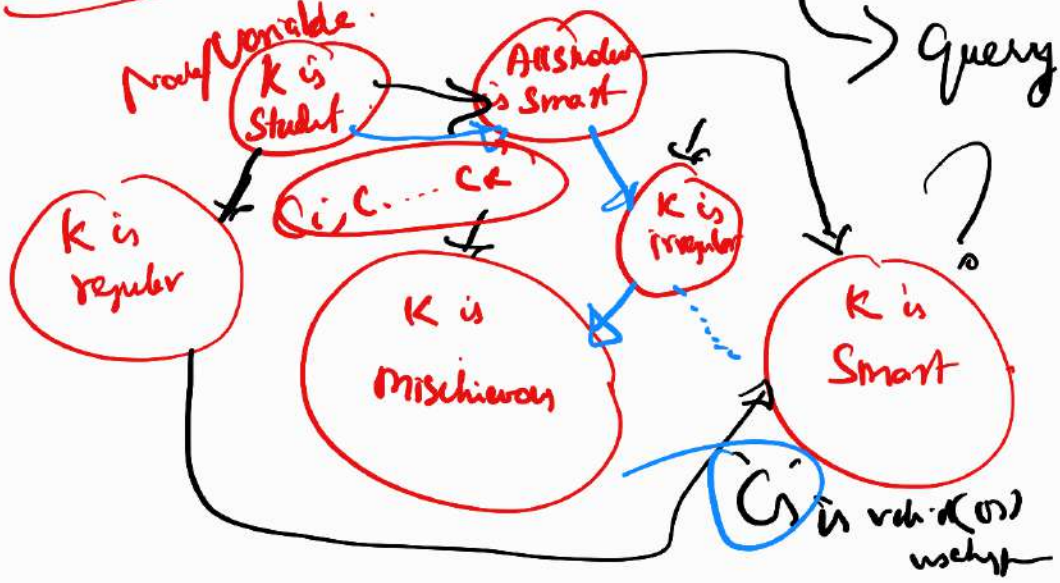
Bayesian N/ws:

$$P(K \mid S, M)$$

Evidence

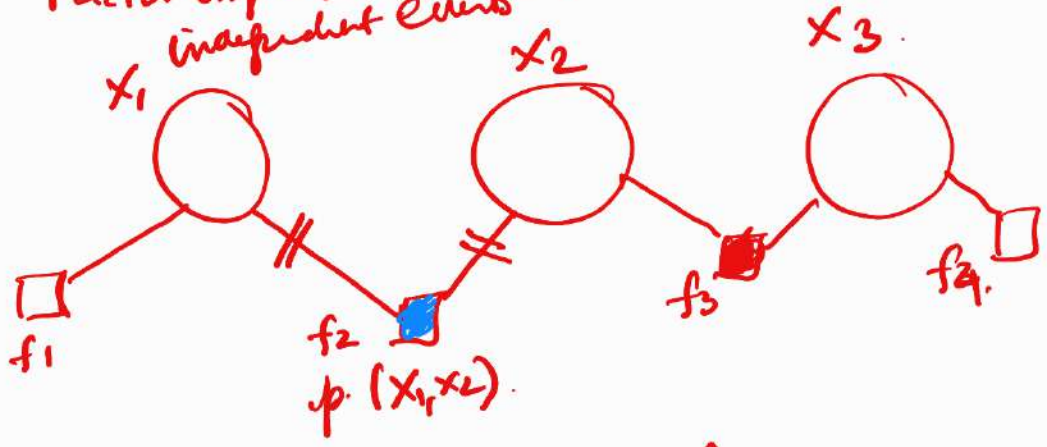
Query

Uncertainties:



Factor Graphs w.r.to Bayes Net

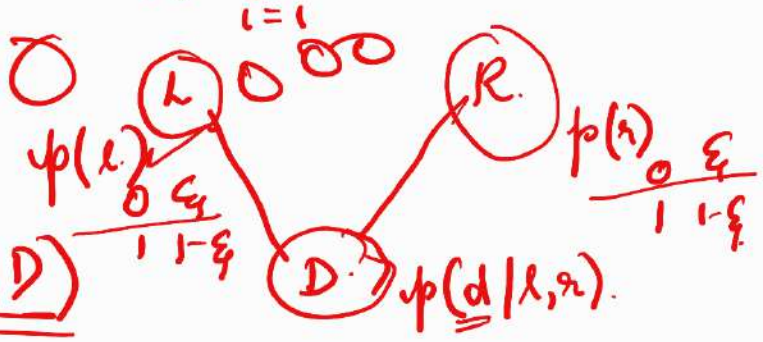
Factor Graphs (Bayes Belief N/w):-
independent events



Given $X\{x_1 \ldots\ldots x_n\}$.

↳ the value of factor $f(x_i) \geq 0$.

$i = 1$



$P(L, R, D)$